

Mathematical Foundations of AI (July-Dec/26)
Problem Sheet - 2

1. In many experiments, one collects the value of a parameter at various instances of time. Let y_i be the value of the parameter y at time x_i . Suppose we wish to construct the best *linear* approximation ($y = mx + b$) to the data in the sense that we wish to minimize the mean square error. Here error is measured vertically rather than perpendicular to the line. Develop formulas for m and b to minimize the mean square error of the points $\{(x_i, y_i) : 1 \leq i \leq n\}$ to the line $y = mx + b$.
2. Manually find the best fit lines (not subspaces which must contain the origin) through the points in the sets below. Best fit means minimise the perpendicular distance. Subtract the center of gravity of the points in the set from each of the points in the set and find the best fit line for the resulting points. Does the best fit line for the original data go through the origin?
 - (a) (4,4) (6,2)
 - (b) (4,2) (4,4) (6,2) (6,4)
 - (c) (3,2.5) (3,5) (5,1) (5,3.5)
3. Manually find the left and right-singular vectors, the singular values, and the SVD decomposition of the following matrices.

$$(a) \quad M = \begin{bmatrix} 1 & 1 \\ 0 & 3 \\ 3 & 0 \end{bmatrix}; \quad (b) \quad M = \begin{bmatrix} 0 & 2 \\ 2 & 0 \\ 1 & 3 \\ 3 & 1 \end{bmatrix};$$

4. Compute the distance between $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$ using (a) $\langle x, y \rangle = x^T y$, and (b) $\langle x, y \rangle = x^T A y$ where $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

5. Compute the angle between $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ using (a) $\langle x, y \rangle = x^T y$, and (b) $\langle x, y \rangle = x^T B y$ where $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$.
6. Compute all eigenspaces of $A = \begin{bmatrix} .0 & -1 & 1 & 1 \\ -1 & 1 & -2 & 3 \\ 2 & -1 & 0 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$
7. Find the best rank-1 approximation of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \end{bmatrix}$ using SVD.
8. Find the best rank-2 approximation of $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \\ 3 & 4 & 5 & 1 & 2 \\ 4 & 5 & 1 & 2 & 3 \end{bmatrix}$ using SVD.